

Mukul Atreya

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EDUCATION

Sam Houston State University

Huntsville, TX

Master of Science in Statistics and Data Science | GPA: 4.0/4.0

August 2025 – May 2027(Expected)

- Relevant Coursework: AI, Applied Probability, Time Series, Bayesian Statistics, Data Analytics, Biostatistics

Institute of Engineering Pulchowk Engineering Campus

Nepal

Bachelor of Science in Computer Engineering

November 2018 – April 2023

EXPERIENCE

Sam Houston State University

Huntsville, TX

Graduate Assistant

August 2025 – Present

- Develop statistical models in R/Python for faculty research projects supporting outcome analysis.
- Deliver classroom lectures on elementary statistics to 20+ undergraduate students, reinforcing core concepts.
- Mentor 30+ undergraduates annually at the Academic Success Center, helping improve academic performance.
- Guide 40+ students in SPSS and statistical analysis during lab sessions.

G-MANA

Remote, Israel

Software Engineer

September 2022 – July 2025

- Engineered high-performance ad-serving system processing 1M+ concurrent viewers with 90%+ fill rate
- Optimized FFmpeg video transcoding pipeline by (1 min to 8 sec) through parallel processing and GPU utilization
- Led full-stack migration from JavaScript to TypeScript across 10+ microservices, reducing runtime errors by 30%.
- Redesigned CI/CD pipeline with parallel builds and automated testing, cutting deployment time by 50%.
- Migrated infrastructure from on-prem to AWS/Hetzner Cloud, reducing costs by 40% while maintaining SLA.
- Implemented real-time monitoring with Elasticsearch + Grafana, reducing system downtime.
- Conducted load testing with JMeter simulating 1M+ users to identify and fix performance bottlenecks.

PROJECTS

Probability of Default Credit Risk Scoring | Logistic Regression, Python, scikit-learn, pandas, matplotlib

- Built a credit risk prediction model to estimate default probability for 150,000+ borrowers.
- Executed exploratory analysis, correlation, feature scaling on 10+ financial and demographic predictors.
- Achieved an AUC of 0.79 on the validation set, demonstrating robust classification.
- Translated predicted probabilities into credit scores with log-odds transformation, enabling credit risk assessment.

Macroeconomic GDP Growth Prediction using Statistical Modeling | R, glmnet, MASS

- Developed GDP growth models using long-run macroeconomic time series and lagged predictors.
- Engineered features (growth-rates, yield-curve measures, and lagged variables) to capture temporal dynamics.
- Achieved Adjusted R^2 of 0.89 using LASSO regularization to identify key predictors.

N-Body Gravity Simulation using CUDA | C++, CUDA, NVIDIA Nsight

- Parallelized $O(N^2)$ N-body physics simulation using CUDA kernels, achieving 8x speedup over CPU.
- Optimized memory throughput using Unified Memory and Prefetching to minimize page faults.
- Eliminated race conditions by implementing separate kernels for force calculation and position integration.
- Profiled performance bottlenecks using Nvidia Nsight Systems to maximize Streaming Multiprocessor (SM) occupancy.

FAANG Resume Builder | JavaScript, React, LaTeX

- Designed an end-to-end resume creation platform generating professional ATS-optimized LaTeX resume pdf.
- Built an automated export-to-email workflow to deliver resume, reducing manual download and sharing steps.
- Streamlined the user experience by building a responsive UI components
- Addressed non-blocking workflow to maintain UI responsiveness during compute-intensive compilation.

TECHNICAL SKILLS

Languages & Frameworks: Python, R, SAS, C/C++, JavaScript, FFmpeg, Node.js, Vue.js

Machine Learning & Data Science: LangChain, Pinecone, PyTorch, TensorFlow, Scikit-learn, Pandas, Grafana

Databases: Postgres, MySQL, MongoDB, Redis, Elasticsearch, SQL

Cloud & DevOps: AWS, Kubernetes, Docker, Hetzner Cloud, Jenkins